UNIT 2 · TOPIC 11(A)

Construction of Farm Equipment — Types, Materials and Heat Treatment

Syllabus topic: Construction and operation of farm equipment

The Unit 2 source notes had no material for this syllabus topic. This file is written from the ICAR eCourse "Farm Machinery and Equipment I" (PAU/ICAR); two slips in that course text are corrected and marked below.

Terminology

Term	Meaning
Tool	A simple mechanical or electro-mechanical device for a task; also an individual working element of a machine (disc, shovel, cutting blade)
Implement	Equipment with no moving or driven parts (plough, cultivator, harrow) — a rigid structure carrying working tools
Machine	A combination of rigid bodies with linkages executing specified motions to do a task
Equipment	General term for any tool, implement, machine or gadget
Device	Any mechanical contrivance offering mechanical advantage
Gadget	A small, relatively simple tool or machine

How equipment is attached and carried

Type	Construction / support
Pull-type (trailed)	Pulled and guided from a single hitch point; not completely supported by the tractor — has its own wheels
Mounted	Hitched through the three-point linkage and fully supported by the tractor when raised; the linkage gives rotational stability and depth/height control
Semi-mounted	Attached through a (nearly) horizontal hinge axis; partly supported by the tractor at least in transport, never fully; large units use rear/middle wheels and remote hydraulic cylinders
Self-propelled	The power unit is an integral part of the machine

- **Mounted — advantages:** cheaper than equivalent pull-type (no support wheels or their structure); a single depth/height control on the tractor; better manoeuvrability and visibility; easier transport; draft sensing and vertical load transfer to aid traction; easy attaching and detaching — helped by the standardised three-point hitch and quick-attach couplers.
- **Mounted — limitations:** the carrying capacity of the tractor chassis, and transport stability.
- **Self-propelled — advantages:** greater flexibility and manoeuvrability, better visibility and operator control, improved mobility, and lower losses when the cutting unit is ahead of the machine. **Disadvantage:** higher investment — it needs more annual use to compete with a pull-type machine.

Materials of construction

Strength, durability and service life depend largely on the material. Design each part to use the lowest-cost material that performs satisfactorily and gives adequate life; costly materials and treatments are a remedy for design deficiencies, not a starting point.

Group	Members
Metallic — ferrous	Iron (cast iron, wrought iron); steel and steel alloys
Metallic — non-ferrous	Copper, zinc, lead, tin, aluminium; brass (Cu + Zn), bronze (Cu + Sn), white metal/Babbitt, solder (Sn + Pb)
Non-metallic	Wood (timber), rubber, leather, plastics, fibreglass, Teflon

Steel (by carbon, eCourse)	Carbon	Uses in farm machines
Low-carbon	up to 0.25%	Frames and most other members — easy to bend, forge and shear
Medium-carbon	0.25–0.50%	Members needing more strength and hardness — shafts, connecting rods
High-carbon	0.50–1.2%	Very hard — tools, ball and roller bearings, cutting tools

Carbon boundaries differ slightly between texts (Unit 8, Topic 1a uses 0.3% and 0.6%); the trend is what matters — more carbon, more hardness, less ductility.

Alloy steel	Composition (eCourse)	Typical parts
Manganese (Hadfield)	11–14% Mn, 0.8–1.5% C — extreme hardness and ductility; cast and ground	Parts under severe wear, e.g., feed grinders
Nickel	2–5% Ni, 0.10–0.5% C — strong, tough, ductile	Parts under repeated shock
Vanadium	~0.20% V — higher tensile strength and elasticity	—
Chrome–vanadium	0.5–1.5% Cr, 0.15–0.3% V, 0.15–1.10% C	Springs, shafts, gears, pins, forgings
Tungsten	3–18% W, 0.2–1.5% C	Dies, high-speed cutting tools
Chrome	0.5–2% Cr, 0.10–1.5% C (14–18% Cr gives stainless grades)	Balls, rollers and races of bearings
Chrome–nickel	0.3–2% Cr, 1.0–4.0% Ni, 0.10–0.60% C	Gears, forgings, crankshafts, connecting rods
Stainless	16–18% Cr, 7–8% Ni, < 0.15% C	Corrosion-resistant parts
Boron	Trace boron raises hardenability	Axles, wheel spindles, steering-knuckle arms, cap screws, studs

Correction: the eCourse table lists boron steel as containing "2.3%" boron. Boron steels contain only trace boron — about 0.0005–0.003% (Unit 8, Topic 1a); that tiny addition is what raises hardenability.

Cast irons (carbon 2.4–4.5%)

Cast iron	Structure	Farm-machine examples
Grey	Carbon as graphite flakes — grey, crystalline fracture	Housings, blocks
White	Carbon chemically combined — white fracture, very hard	—
Chilled	Selected surfaces cooled rapidly (white-iron face) over a grey-iron body	Mould boards (one side hardened)
Malleable	White iron annealed at ~900 °C and cooled slowly — tough, strong, machinable	Mower guards, ledger plates, clutch/brake pedals, chain links
Spheroidal-graphite (nodular / ductile)	Magnesium added to the melt gives spheroidal graphite (developed 1949)	Sprockets, gears, plough shares, mower guards, baler-knotter parts

(The eCourse lists the fifth type as "malleable" a second time; its description — magnesium-treated, spheroidal graphite — is SG/ductile iron.)

Non-ferrous metals

Metal / alloy	Composition	Uses
Copper	—	Tubing, wires, windings
Brass	Cu 60–90% + Zn 10–40%	Radiator pipes, brazing rods, fuel-line screens, instrument parts
Bronze	Cu 80–95% + Sn 5–20% (sometimes Zn)	Bushings, springs, pipe fittings, valves, pump pistons, bearings
Babbitt (white metal)	Tin-based: Sn 80–84%, Sb 8–9%, Cu 7–8%	Bearings

Heat treatment of farm-equipment parts

Heat treatment = heating and cooling in a prescribed manner (time, temperature, rates) to give metals the desired properties in the solid state.

Process	What is done	Purpose
Hardening	Steel with > 0.5% C heated to ~850 °C (red) and quenched in water, brine or oil	Maximum hardness — but brittle, so it must be tempered
Tempering	Hardened part reheated below the transformation range (~500 °C, blue colour) and cooled	Removes most brittleness at a small cost in hardness
Full annealing	Heated to ~900 °C (above the transformation range), held, then cooled very slowly in the furnace	Removes forging/casting stresses; done before hardening
Normalising	Heated above the transformation range, then cooled freely in air	Grain refinement, uniform structure
Stress relieving	Heated below the transformation range, then cooled slowly	Removes internal stresses only
Case hardening (pack carburising)	Low-carbon steel packed in charcoal-based compound at ~900 °C for 5–10 h; reheated to ~800 °C and quenched	Hard, wear-resistant case over a tough, shock-resistant core
Nitriding / carbonitriding / cyaniding	Surface enriched with nitrogen (ammonia), carbon + nitrogen, or from a molten cyanide bath	Thin hard surface layer
Induction hardening	High-frequency current heats the surface locally; water quench	Local surface hardening
Flame hardening	Oxy-acetylene flame heats the surface quickly (~900 °C); water quench	Local surface hardening

- The faster the cooling, the harder and more brittle the steel. Quench severity: **brine > water > oil > air**.
- Plough shares, cultivator points and rake teeth are hardened and tempered under controlled conditions for resistance to abrasion and bending (see also Unit 8, Topic 1c).

Correction: the eCourse states that brine cools more slowly than water. Brine is the more severe (faster) quench; the order above is the standard one.

Example — construction of a mould-board plough bottom

Share type	Construction	Usual material
Slip share	One piece with curved cutting edge — simple, common; whole share replaced when worn	High-carbon steel, soft-centre steel, cast iron
Slip-nose share	Point is a small detachable piece — economical to renew	Cast iron
Shin share	Slip share with an extension (shin) along the mould board	High-carbon steel
Bar-point share	Point formed by an adjustable, replaceable bar that also serves as landside	High-carbon steel

Share material	Character / where used
Cast iron	Cheap; brittle — breaks on a sharp blow; can be re-sharpened by grinding when worn
Chilled cast iron	Longer life — does not rust or wear quickly; for sandy and strong soils
Soft-centre steel	Very hard surface over a tough core; long life but costly; for soils that do not stick
Solid (forged high-carbon) steel	Sturdy, withstands shock; for non-abrasive soils

- **Landside:** about 23 cm (where landside pressure is not a factor), 28 cm (normal ploughing), 36 cm with heel (abrasive soils), 50 cm with heel (ploughs without a rear furrow wheel). Material: soft-centre steel, mild steel or cast iron.
- **Frog:** mild steel or cast iron.

Self-check

Question	Answer
Which attachment type is cheapest for the same work, and why?	Mounted — no support wheels or carrying structure; depth control comes from the tractor
Most frames and members of farm machines are made of…	Low-carbon steel
Which cast iron is used for mower guards, ledger plates and chain links?	Malleable cast iron
What does case hardening give a part?	A hard wear-resistant case on a tough core
Why must a hardened share be tempered?	Hardening leaves it brittle; tempering restores toughness against shock
Which share has a replaceable point?	Slip-nose share

Sources: ICAR eCourse "Farm Machinery and Equipment I" — course content by Dr Ritu Dogra (Dept. of Farm Machinery & Power Engg., PAU Ludhiana), reviewed by Dr K. K. Singh (ICAR) — Lessons 2 (materials of construction), 3 (heat treatment), 5–6 (field capacity, field patterns), 8–9 (mould-board plough construction, adjustments, operation) and 24 (calibration); published as an AgriMoon e-book.